

## Program 5:

Project Structure:

Pr5

->Dockerfile

->docker-compose.yaml

### Step 1: Create docker-compose.yaml

version: '3.8'

# <https://kubernetes.io/docs/concepts/services-networking/service/>

services:

jenkins:

build:

container\_name: jenkins-ci

ports:

- "8080:8080"

- "50000:50000"

volumes:

- jenkins\_home:/var/jenkins\_home

- /var/run/docker.sock:/var/run/docker.sock

volumes:

jenkins\_home:

### Step 2: Create Dockerfile

FROM jenkins/jenkins:its

USER root

RUN apt-get update \

&& apt-get install -y docker.io \

&& rm -rf /var/lib/apt/lists/\*

# Add 'jenkins' user to the 'docker' group

RUN usermod -aG docker jenkins

USER jenkins

### Step 3: Build and run jenkins

```
jenkins:
  build: .
  container_name: jenkins-ci
  ports:
    - "8080:8080"
    - "50000:50000"
  volumes:
    - jenkins_home:/var/jenkins_home
    - /var/run/docker.sock:/var/run/docker.sock

volumes:
  jenkins_home:
}
}
) docker compose build
WARN[0000] /home/kumars/Development/devops/docker/docker1/docker-compose.yaml: the attribute `version` is
obsolete, it will be ignored, please remove it to avoid potential confusion
[+] Building 10.0s (3/4)
=> [internal] load local bake definitions                                0.0s
=> => reading from stdin 548B                                           0.0s
=> [internal] load build definition from Dockerfile                     0.0s
=> => transferring dockerfile: 248B                                     0.0s
=> [internal] load metadata for docker.io/jenkins/jenkins:lts         10.0s
=> [auth] jenkins/jenkins:pull token for registry-1.docker.io         0.0s

=> [internal] load .dockerignore                                         0.0s
=> => transferring context: 2B                                           0.0s
=> [1/3] FROM docker.io/jenkins/jenkins:lts@sha256:1e407df1a9bcb122cc4d71d4ce44d2e3b2be31af004c 173.7s
=> => resolve docker.io/jenkins/jenkins:lts@sha256:1e407df1a9bcb122cc4d71d4ce44d2e3b2be31af004c15 0.0s
=> => sha256:d2519f76d4c6527e03084f2742da3ae15b0c5097d464be4dd93de4f573749313 392B / 392B 0.9s
=> => sha256:986be6969b0a04975a2e559b30a96d766bf360d0b340b2789b611688ba8bbeea 1.29kB / 1.29kB 1.4s
=> => sha256:3fb70446889ebcc58e91f43f90aafa314798941c318ce016ea62068a3de53927 2.23kB / 2.23kB 2.1s
=> => sha256:d7dd2e63023897146ec4b913a460382968fbbd1cc3682e36f650727a1bfd6899 72.05MB / 72.05MB 171.4s
=> => sha256:a9230f7c9c31780c6b1dc4f2aa7e1073df2fdf88e9c009f6874e7221acb06245 6.29MB / 6.29MB 14.8s
=> => sha256:687044b042c5d6d57e2ce207c000ee159f2382cb84cb8d80688c35e7f1ed8627 5.24MB / 95.32MB 172.3s
=> => sha256:501deb283b89e30fd3f1f4e27a635bcc1739485e8ba805034552e3d4eaac0b98 189B / 189B 1.1s
=> => sha256:e0e73f34dee93322ae4db1d44250aab1f61011a2b7ec9549ff1f6df342c31168 183B / 183B 0.3s
=> => sha256:68cdef48009293f6a12f5c50b91fcfe4194bd6293334d8a31a45b1ed63cb91a5 1.23kB / 1.23kB 0.8s
=> => sha256:2f1ecb09fa362ebe652038ecbfcc64a84b042991e9dedb00059a4127c0121471 5.24MB / 5.24MB 40.7s
=> => sha256:5687d4df0728e3558596e7d718163b2e812a00cf5dd2624459c3315dce34b3cc 3.15MB / 65.08MB 158.0s
^C%
) docker compose up
WARN[0000] /home/kumars/Development/devops/docker/docker1/docker-compose.yaml: the attribute `version` is
obsolete, it will be ignored, please remove it to avoid potential confusion
[+] Building 10.4s (3/3) FINISHED
=> [internal] load local bake definitions                                0.0s
=> => reading from stdin 548B                                           0.0s
=> [internal] load build definition from Dockerfile                     0.0s
=> => transferring dockerfile: 248B                                     0.0s
```

## Step 4: List docker images

```
$ sudo docker container ls -a
```

| CONTAINER ID | IMAGE        | COMMAND                  | CREATED                | STATUS                    |
|--------------|--------------|--------------------------|------------------------|---------------------------|
| ca0bfa73393  | pr5-jenkins  | "/usr/bin/tini -- /u..." | Less than a second ago | Up Less than a second ago |
| 802603091e9  | program4     | "python app.py"          | 2 weeks ago            | Exited (0) 2 weeks ago    |
| eb15eb68d40  | 3c45199ff050 | "flask app.py"           | 2 weeks ago            | Exited (2) 2 weeks ago    |
| 46c2302f9c7  | program-4    | "flask app.py"           | 2 weeks ago            | Exited (2) 2 weeks ago    |
| ldcf51ecd4e  | 57f1863e1d53 | "flask run"              | 2 weeks ago            | Exited (0) 2 weeks ago    |
| f392ffdf571  | program3     | "python app.py"          | 2 weeks ago            | Exited (0) 2 weeks ago    |
| fa1bc3b7841  | 52d6d6f80aea | "python app.py"          | 2 weeks ago            | Exited (0) 2 weeks ago    |
| 0c42265eda5  | 52d6d6f80aea | "python app.py"          | 2 weeks ago            | Exited (0) 2 weeks ago    |
| 8a518be1f50  | 7a5ec8552c52 | "python app.py"          | 2 weeks ago            | Exited (0) 2 weeks ago    |
| e8fdd8fce8f  | 5cdc8ed4f59b | "docker-entrypoint.s..." | 2 weeks ago            | Exited (0) 2 weeks ago    |
| 8481cea2cbc  | 5cdc8ed4f59b | "docker-entrypoint.s..." | 2 weeks ago            | Exited (0) 2 weeks ago    |

## Step 5 :Open localhost:8080

localhost:8080

Getting Started

# Customize Jenkins

Plugins extend Jenkins with additional features to support many different needs

**Install suggested plugins**

Install plugins the Jenkins community finds most useful.

**Select plugins to install**

Select and install plugins most suitable for your needs.